



Pokhara University
Everest Engineering College
First Internal Assessment Practice Sets
Fall-2025

Level: Bachelor **F.M. 100**
Program: BE Software **P.M. 45**
Section: A/B **Time: 3hrs**
Subject: System Programming

Attempt all the questions.

1	a) Define System Software. Explain the architecture of SIC Machine with important features.	8
	b) How does RISC machine differ from SIC machine. Briefly state major differences between RISC and CISC architecture.	7
2	a) Write about instruction formats, addressing mode and registers of SIC/XE architecture	8
	b) How does CISC machine differ from SIC machine. Briefly state major differences between RISC and CISC architecture.	7
3	a) Define SIC assembler. Explain the basic assembler functions along with a example.	8
	b) Write a program in SIC to perform arithmetic operation between two integers	7

4	a) Define SIC Assembler. Briefly write the steps (functions) to translate source program to object code.	8																																																												
	b) Write a program in SIC to find maximum between two integers.	7																																																												
5	a) Briefly explain machine dependent features of an assembler.	7																																																												
	b) How does machine dependent features of assembler differ from machine independent features. Write and explain about one pass assembler.	8																																																												
6	a) Consider the following assembly language program: i. Fill Column for location counter ii. Create object code column with object codes. <table><tr><th>Line</th><th>Symbol</th><th>Opcode</th><th>Exp</th></tr><tr><td>10</td><td>STRCPY</td><td>START</td><td>1000</td></tr><tr><td>20</td><td>FIRST</td><td>LDX</td><td>ZERO</td></tr><tr><td>30</td><td>MOVECH</td><td>LDCH</td><td>STR1, X</td></tr><tr><td>40</td><td></td><td>STCH</td><td>STR2, X</td></tr><tr><td>50</td><td></td><td>TIX</td><td>ELEVEN</td></tr><tr><td>60</td><td></td><td>JLT</td><td>MOVECH</td></tr><tr><td>70</td><td>STR1</td><td>BYTE</td><td>C'ABCD'</td></tr><tr><td>80</td><td>STR2</td><td>RESB</td><td>11</td></tr><tr><td>90</td><td>ZERO</td><td>WORD</td><td>0</td></tr><tr><td>100</td><td>ELEVEN</td><td>WORD</td><td>11</td></tr><tr><td>110</td><td></td><td>END</td><td>FIRST</td></tr></table> <table><tr><th>Mnemonics</th><th>Opcode</th></tr><tr><td>LDCH</td><td>50</td></tr><tr><td>LDX</td><td>04</td></tr><tr><td>STCH</td><td>54</td></tr><tr><td>JLT</td><td>38</td></tr><tr><td>TIX</td><td>2C</td></tr></table>	Line	Symbol	Opcode	Exp	10	STRCPY	START	1000	20	FIRST	LDX	ZERO	30	MOVECH	LDCH	STR1, X	40		STCH	STR2, X	50		TIX	ELEVEN	60		JLT	MOVECH	70	STR1	BYTE	C'ABCD'	80	STR2	RESB	11	90	ZERO	WORD	0	100	ELEVEN	WORD	11	110		END	FIRST	Mnemonics	Opcode	LDCH	50	LDX	04	STCH	54	JLT	38	TIX	2C	8
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	b) Explain the major differences between One pass assembler and multi pass assembler.	7																																																												
7	Write short notes on: (Any two) a) Symbol defining statements and expressions b) Program Relocation c) Basic Loader Functions	2*5 =10																																																												

Good Luck